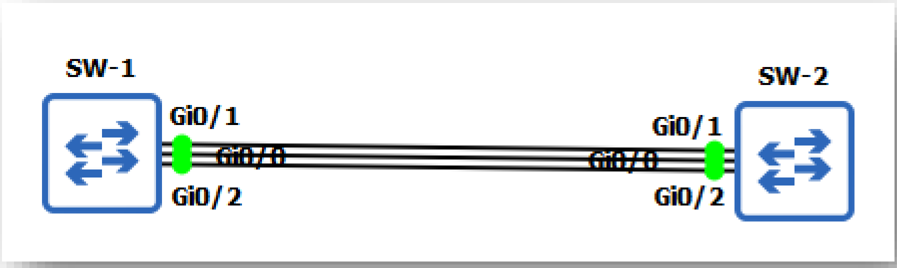
Layer-3 EtherChannel



Task

1. Configure the SW-1 with IP 150.150.150.1 and SW-2 with IP 150.150.150.2 for device management and hostnames.
2. Configure Layer-3 EtherChannel (LACP) with Group number 10
3. Configure IP address 30.0.0.0/30 on Port-Channel 10

Task-1: Configure Hostname on switches as SW-1 and SW-2. Configure the SW-1 with IP 150.150.150.1 and SW-2 with IP 150.150.150.2 for device management.

SW-1 Configuration:
Switch#config t
Switch(config)#hostname SW-1
SW-1(config)#no ip domain-lookup
SW-1(config)#int vlan1
SW-1(config-if)#ip address 150.150.150.1 255.255.255.0
SW-1(config-if)#no shut
SW-1(config-if)#exit
SW-1(config)#exit
SW-1#

SW-2 Configuration:
Switch#config t
Switch(config)#hostname SW-2
SW-2(config)#no ip domain-lookup
SW-2(config)#int vlan1
SW-2(config-if)#ip address 150.150.150.2 255.255.255.0
SW-2(config-if)#no shut
SW-2(config-if)#exit
SW-2(config)#exit
SW-2#

✓ **Verification & Testing:**

SW-1#sh spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID   Priority   32769

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```
         Address     0c4d.da1e.4800
         Cost        4
         Port        1 (GigabitEthernet0/0)
         Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
         Address     0c4d.daf9.9600
         Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
         Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Altn BLK 4         128.2    P2p
Gi0/2               Altn BLK 4         128.3    P2p
Gi0/3               Desg FWD 4         128.4    P2p
Gi1/0               Desg FWD 4         128.5    P2p
Gi1/1               Desg FWD 4         128.6    P2p
Gi1/2               Desg FWD 4         128.7    P2p
Gi1/3               Desg FWD 4         128.8    P2p
Gi2/0               Desg FWD 4         128.9    P2p
Gi2/1               Desg FWD 4         128.10   P2p
Gi2/2               Desg FWD 4         128.11   P2p
Gi2/3               Desg FWD 4         128.12   P2p
Gi3/0               Desg FWD 4         128.13   P2p
Gi3/1               Desg FWD 4         128.14   P2p
Gi3/2               Desg FWD 4         128.15   P2p
Gi3/3               Desg FWD 4         128.16   P2p
```

SW-2#sh spanning-tree

```
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
         Address     0c4d.da1e.4800
         This bridge is the root
         Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
         Address     0c4d.da1e.4800
         Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
         Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Desg FWD 4         128.1    P2p
Gi0/1               Desg FWD 4         128.2    P2p
Gi0/2               Desg FWD 4         128.3    P2p
Gi0/3               Desg FWD 4         128.4    P2p
Gi1/0               Desg FWD 4         128.5    P2p
Gi1/1               Desg FWD 4         128.6    P2p
```

```
Gi1/2          Desg FWD 4     128.7    P2p
Gi1/3          Desg FWD 4     128.8    P2p
Gi2/0          Desg FWD 4     128.9    P2p
Gi2/1          Desg FWD 4     128.10   P2p
Gi2/2          Desg FWD 4     128.11   P2p
Gi2/3          Desg FWD 4     128.12   P2p
Gi3/0          Desg FWD 4     128.13   P2p
Gi3/1          Desg FWD 4     128.14   P2p
Gi3/2          Desg FWD 4     128.15   P2p
Gi3/3          Desg FWD 4     128.16   P2p
```

SW-1#sh int gi0/0 switchport
Name: Gi0/0
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: static access
Administrative Trunking Encapsulation: negotiate
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled

Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk Native VLAN tagging: enabled
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk associations: none
Administrative private-vlan trunk mappings: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL

Protected: false
Appliance trust: none

SW-1#sh int gi0/1 switchport | section Switchport
Switchport: Enabled

SW-1#sh int gi0/2 switchport | section Switchport
Switchport: Enabled

SW-2#sh int gi0/0 switchport | section Switchport
Switchport: Enabled

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SW-2#sh int gi0/1 switchport | section Switchport
Switchport: Enabled

SW-2#sh int gi0/2 switchport | section Switchport
Switchport: Enabled

Task-2: Configure Layer-3 EtherChannel (LACP) with Group number 10

LACP Modes:

Mode	Description
active	Places a port into an active negotiating state in which the port starts negotiations with other ports by sending LACP packets.
passive	Places a port into a passive negotiating state in which the port responds to LACP packets that it receives, but does not start LACP packet negotiation. This setting minimizes the transmission of LACP packets.

- A port in the **active** mode can form an EtherChannel with another port that is in the **active** or **passive** mode.
- A port in the **passive** mode cannot form an EtherChannel with another port that is also in the **passive** mode because neither port starts LACP negotiation.

SW-1 Configuration:

SW-1#config t
SW-1(config)#interface range gi0/0 - 2
SW-1(config-if-range)#no switchport
SW-1(config-if-range)#no ip address
SW-1(config-if-range)#channel-group 10 mode active
Creating a port-channel interface Port-channel 10
SW-1(config-if-range)#exit
SW-1(config)#exit
SW-1#

SW-2 Configuration:

SW-2#config t
SW-2(config)#interface range gi0/0 - 2
SW-2(config-if-range)#no switchport
SW-2(config-if-range)#no ip address
SW-2(config-if-range)#channel-group 10 mode active
Creating a port-channel interface Port-channel 10
SW-2(config-if-range)#exit
SW-2(config)#exit
SW-2#

Task-3: Configure IP address on Port-Channel 10

SW-1 Configuration:

```
SW-1#config t
SW-1(config)#int port-channel 10
SW-1(config-if)#no switchport
SW-1(config-if)#ip address 30.0.0.1 255.255.255.252
SW-1(config-if)#exit
SW-1(config)#exit
SW-1#
```

SW-2 Configuration:

```
SW-2#config t
SW-2(config)#int port-channel 10
SW-2(config-if)#no switchport
SW-2(config-if)#ip address 30.0.0.2 255.255.255.252
SW-2(config-if)#exit
SW-2(config)#exit
SW-2#
```

✓ Verification & Testing:

SW-1#sh ip int brief

```
Interface            IP-Address      OK? Method Status     Protocol
GigabitEthernet0/0   unassigned      YES manual up         up
GigabitEthernet0/1   unassigned      YES manual up         up
GigabitEthernet0/2   unassigned      YES manual up         up
GigabitEthernet0/3   unassigned      YES unset  down       down
GigabitEthernet1/0   unassigned      YES unset  down       down
GigabitEthernet1/1   unassigned      YES unset  down       down
GigabitEthernet1/2   unassigned      YES unset  down       down
GigabitEthernet1/3   unassigned      YES unset  down       down
GigabitEthernet2/0   unassigned      YES unset  down       down
GigabitEthernet2/1   unassigned      YES unset  down       down
GigabitEthernet2/2   unassigned      YES unset  down       down
GigabitEthernet2/3   unassigned      YES unset  down       down
GigabitEthernet3/0   unassigned      YES unset  down       down
GigabitEthernet3/1   unassigned      YES unset  down       down
GigabitEthernet3/2   unassigned      YES unset  down       down
GigabitEthernet3/3   unassigned      YES unset  down       down
Port-channel10       30.0.0.1        YES manual up         up
Vlan1                150.150.150.1   YES manual up         up
```

SW-2#sh ip int brief

```
Interface            IP-Address      OK? Method Status     Protocol
GigabitEthernet0/0   unassigned      YES manual up         up
GigabitEthernet0/1   unassigned      YES manual up         up
```

```
GigabitEthernet0/2    unassigned      YES manual up                    up
GigabitEthernet0/3    unassigned      YES unset  down                  down
GigabitEthernet1/0    unassigned      YES unset  down                  down
GigabitEthernet1/1    unassigned      YES unset  down                  down
GigabitEthernet1/2    unassigned      YES unset  down                  down
GigabitEthernet1/3    unassigned      YES unset  down                  down
GigabitEthernet2/0    unassigned      YES unset  down                  down
GigabitEthernet2/1    unassigned      YES unset  down                  down
GigabitEthernet2/2    unassigned      YES unset  down                  down
GigabitEthernet2/3    unassigned      YES unset  down                  down
GigabitEthernet3/0    unassigned      YES unset  down                  down
GigabitEthernet3/1    unassigned      YES unset  down                  down
GigabitEthernet3/2    unassigned      YES unset  down                  down
GigabitEthernet3/3    unassigned      YES unset  down                  down
Port-channel10        30.0.0.2        YES manual up                    up
Vlan1                 150.150.150.2   YES manual up                    up
```

SW-1#sh int status

```
Port      Name               Status       Vlan       Duplex  Speed Type
Gi0/0                        connected    routed     a-full  auto RJ45
Gi0/1                        connected    routed     a-full  auto RJ45
Gi0/2                        connected    routed     a-full  auto RJ45
Gi0/3                        notconnect   1          a-full  auto RJ45
Gi1/0                        notconnect   1          a-full  auto RJ45
Gi1/1                        notconnect   1          a-full  auto RJ45
Gi1/2                        notconnect   1          a-full  auto RJ45
Gi1/3                        notconnect   1          a-full  auto RJ45
Gi2/0                        notconnect   1          a-full  auto RJ45
Gi2/1                        notconnect   1          a-full  auto RJ45
Gi2/2                        notconnect   1          a-full  auto RJ45
Gi2/3                        notconnect   1          a-full  auto RJ45
Gi3/0                        notconnect   1          a-full  auto RJ45
Gi3/1                        notconnect   1          a-full  auto RJ45
Gi3/2                        notconnect   1          a-full  auto RJ45
Gi3/3                        notconnect   1          a-full  auto RJ45
Po10                         connected    routed     a-full  auto
```

SW-2#sh int status

```
Port      Name               Status       Vlan       Duplex  Speed Type
Gi0/0                        connected    routed     a-full  auto RJ45
Gi0/1                        connected    routed     a-full  auto RJ45
Gi0/2                        connected    routed     a-full  auto RJ45
Gi0/3                        notconnect   1          a-full  auto RJ45
Gi1/0                        notconnect   1          a-full  auto RJ45
Gi1/1                        notconnect   1          a-full  auto RJ45
Gi1/2                        notconnect   1          a-full  auto RJ45
Gi1/3                        notconnect   1          a-full  auto RJ45
Gi2/0                        notconnect   1          a-full  auto RJ45
Gi2/1                        notconnect   1          a-full  auto RJ45
```

```
Gi2/2        notconnect   1        a-full  auto RJ45
Gi2/3        notconnect   1        a-full  auto RJ45
Gi3/0        notconnect   1        a-full  auto RJ45
Gi3/1        notconnect   1        a-full  auto RJ45
Gi3/2        notconnect   1        a-full  auto RJ45
Gi3/3        notconnect   1        a-full  auto RJ45
Po10         connected    routed   a-full  auto
```

SW-1#sh int port-channel 10

```
Port-channel10 is up, line protocol is up (connected)
  Hardware is EtherChannel, address is 0c4d.daf9.8000 (bia 0c4d.daf9.8000)
  Internet address is 30.0.0.1/30
  MTU 1500 bytes, BW 2000000 Kbit/sec, DLY 10 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, Auto-speed, media type is RJ45
  input flow-control is off, output flow-control is unsupported
  Members in this channel: Gi0/1 Gi0/2
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:04, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     1361 packets input, 188108 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     0 input packets with dribble condition detected
     1361 packets output, 193065 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out
SW-1#
```

SW-2#sh int port-channel 10 | section Hardware|Internet

```
  Hardware is EtherChannel, address is 0c4d.da1e.8000 (bia 0c4d.da1e.8000)
  Internet address is 30.0.0.2/30
SW-2#
```

SW-1#sh etherchannel summary

```
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
```


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f - failed to allocate aggregator
M - not in use, minimum links not met
m - not in use, port not aggregated due to minimum links not met
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators: 1

```
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
10     Po10(RU)       LACP      Gi0/0(P)    Gi0/1(P)    Gi0/2(P)
```

SW-2#sh etherchannel summary

Flags: D - down   P - bundled in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3   S - Layer2
U - in use   N - not in use, no aggregation
f - failed to allocate aggregator
M - not in use, minimum links not met
m - not in use, port not aggregated due to minimum links not met
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators: 1

```
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
10     Po10(RU)       LACP      Gi0/0(P)    Gi0/1(P)    Gi0/2(P)
```

SW-1#sh lacp neighbor

Flags: S - Device is requesting Slow LACPDUs
F - Device is requesting Fast LACPDUs
A - Device is in Active mode   P - Device is in Passive mode
Channel group 10 neighbors
Partner's information:

		LACP port			Admin	Oper	Port	Port
Port	Flags	Priority	Dev ID	Age	key	Key	Number	State
Gi0/0	SA	32768	0c4d.da1e.8000	9s	0x0	0xA	0x1	0x3D
Gi0/1	SA	32768	0c4d.da1e.8000	10s	0x0	0xA	0x2	0x3D
Gi0/2	SA	32768	0c4d.da1e.8000	8s	0x0	0xA	0x3	0x3D

SW-2#sh lacp neighbor

```
Flags:  S - Device is requesting Slow LACPDUs
        F - Device is requesting Fast LACPDUs
        A - Device is in Active mode       P - Device is in Passive mode

Channel group 10 neighbors
Partner's information:

                  LACP port                        Admin  Oper  Port   Port
Port    Flags   Priority      Dev ID          Age    key    Key    Number  State
Gi0/0     SA      32768    0c4d.daf9.8000   6s    0x0    0xA    0x1        0x3D
Gi0/1     SA      32768    0c4d.daf9.8000   5s    0x0    0xA    0x2        0x3D
Gi0/2     SA      32768    0c4d.daf9.8000  10s    0x0    0xA    0x3        0x3D
```

SW-1#sh etherchannel protocol

```
                Channel-group listing: Group: 10
-----------------------------------------------------------
Protocol:  LACP
```

SW-2#sh etherchannel protocol

```
                Channel-group listing: Group: 10
-----------------------------------------------------------
Protocol:  LACP
```

SW-1#sh etherchannel port-channel

```
                Channel-group listing:
                ----------------------
Group: 10
----------
                Port-channels in the group:
                ---------------------------
Port-channel: Po10    (Primary Aggregator)
------------

Age of the Port-channel          = 0d:00h:13m:57s
Logical slot/port                = 16/0          Number of ports = 3
HotStandBy port                  = null
Passive port list                = Gi0/0 Gi0/1 Gi0/2
Port state                       = Port-channel L3-Ag Ag-Inuse
Protocol                         =   LACP
Port security        = Disabled

Ports in the Port-channel:

Index   Load   Port    EC state        No of bits
------+------+------+------------------+-----------
  0       00     Gi0/0    Active            0
  0       00     Gi0/1    Active            0
  0       00     Gi0/2    Active            0

Time since last port bundled:           0d:00h:01m:25s    Gi0/1
```

Time since last port Un-bundled: 0d:00h:02m:36s   Gi0/1

SW-2#sh etherchannel port-channel

Channel-group listing:

Group: 10

Port-channels in the group:

Port-channel: Po10   (Primary Aggregator)

Age of the Port-channel = 0d:00h:13m:45s
Logical slot/port = 16/0   Number of ports = 3
HotStandBy port = null
Passive port list = Gi0/0 Gi0/1 Gi0/2
Port state = Port-channel L3-Ag Ag-Inuse
Protocol = LACP
Port security = Disabled

Ports in the Port-channel:

Index	Load	Port	EC state	No of bits
0	00	Gi0/0	Active	0
0	00	Gi0/1	Active	0
0	00	Gi0/2	Active	0

Time since last port bundled: 0d:00h:01m:21s   Gi0/0
Time since last port Un-bundled: 0d:00h:02m:30s   Gi0/0

SW-1#sh etherchannel load-balance
EtherChannel Load-Balancing Configuration: src-dst-ip

EtherChannel Load-Balancing Addresses Used Per-Protocol:
Non-IP: Source XOR Destination MAC address
IPv4: Source XOR Destination IP address
IPv6: Source XOR Destination IP address
SW-2#sh etherchannel load-balance
EtherChannel Load-Balancing Configuration: src-dst-ip

EtherChannel Load-Balancing Addresses Used Per-Protocol:
Non-IP: Source XOR Destination MAC address
IPv4: Source XOR Destination IP address
IPv6: Source XOR Destination IP address

SW-1#ping 30.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/4 ms

SW-2#ping 30.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/4 ms

SW-1#sh ip route | begin Gateway
Gateway of last resort is not set

```
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/30 is directly connected, Port-channel10
L        30.0.0.1/32 is directly connected, Port-channel10
      150.150.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        150.150.150.0/24 is directly connected, Vlan1
L        150.150.150.1/32 is directly connected, Vlan1
SW-1#
```

SW-2#sh ip route | begin Gateway
Gateway of last resort is not set

```
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/30 is directly connected, Port-channel10
L        30.0.0.2/32 is directly connected, Port-channel10
      150.150.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        150.150.150.0/24 is directly connected, Vlan1
L        150.150.150.2/32 is directly connected, Vlan1
SW-2#
```


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Important Commands:

sh spanning-tree
sh lacp neighbor
sh etherchannel protocol
sh etherchannel summary
sh etherchannel port-channel
sh int port-channel 10
sh etherchannel load-balance
sh ip route | begin Gateway